

From pollen images to category predictions

A working baseline—and clearer questions for expert review.

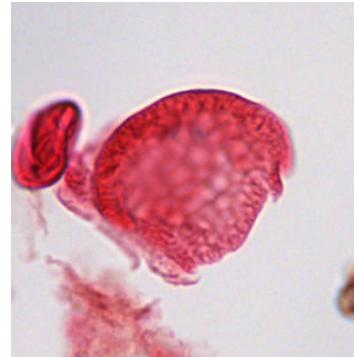
2,283

labeled specimens

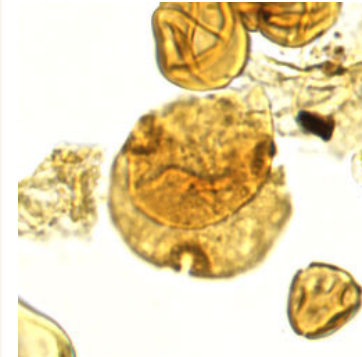
32

priority categories

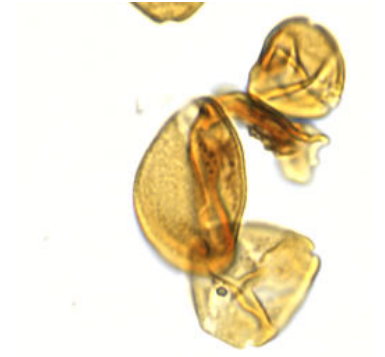
Prepared images
Compared classifiers
Saved results for review



Bisaccate



Bombacacidites sp.

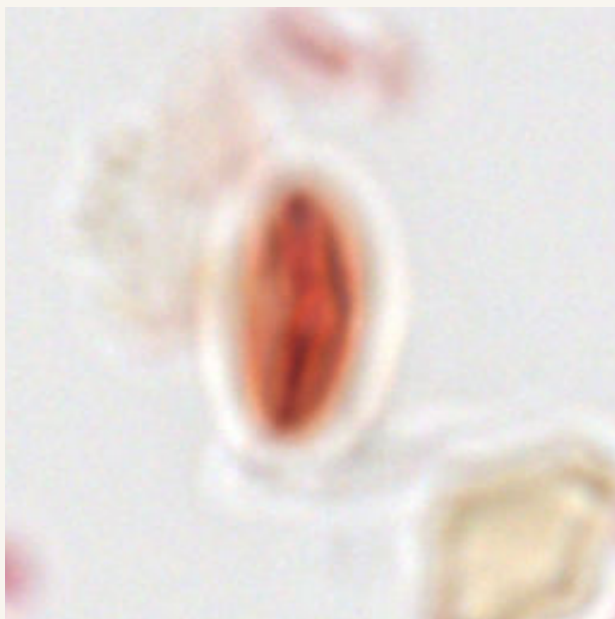


Arecipites sp.

Real specimen crops • expert category labels

A clearer view helps the model identify pollen

Original focal plane



Selected sharper plane



Unseen-slide accuracy

32% → 38%

Same Swin model settings.
Only the selected focal
plane changed.

Same specimen and crop • expert label: Quercus sp.

Two questions, two ways to check performance

Unseen slides

Can it work on a different slide?

LEARN FROM

Slide A

Slide B

CHECK ON

Slide C

Slide D

Shared slides

Can it identify more grains on familiar slides?

LEARN FROM

A · grain 1

B · grain 1

CHECK ON

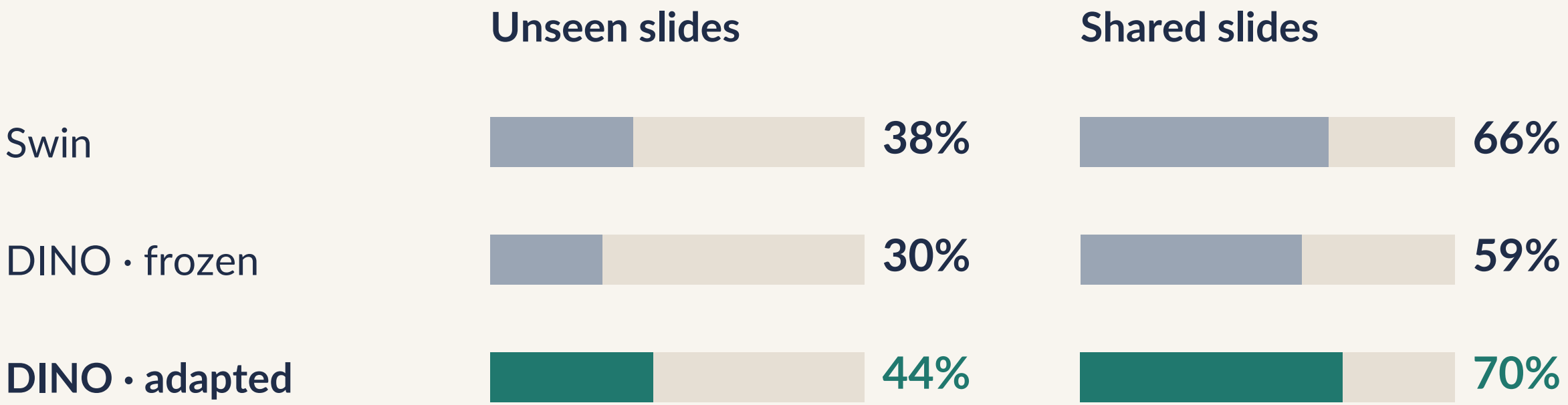
A · grain 2

B · grain 2

Different specimens in every set. Final test slides remain reserved.

Adapting DINO gives our strongest results so far

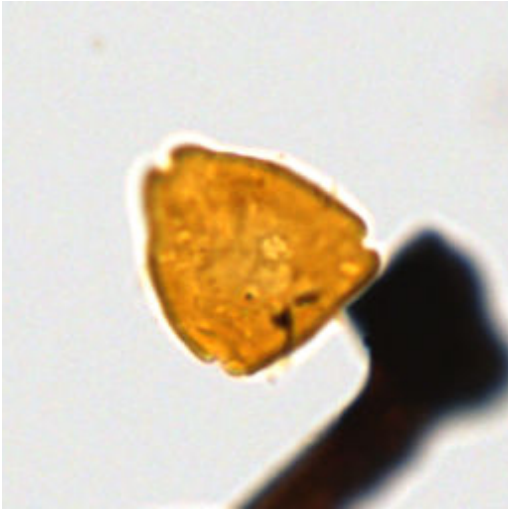
Validation accuracy · identifying categories in prepared specimen crops



Adapted = trained the final two DINO blocks and its category classifier.

Single-run validation comparisons; final test untouched. These are not whole-slide detection results.

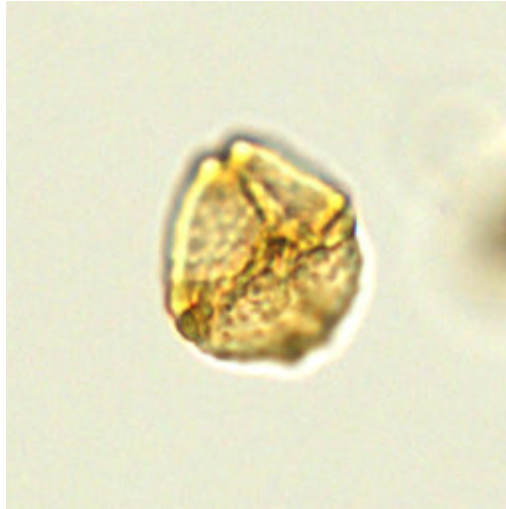
Which differences matter to an expert?



EXPERT LABEL

**Momipites
wyomingensis**

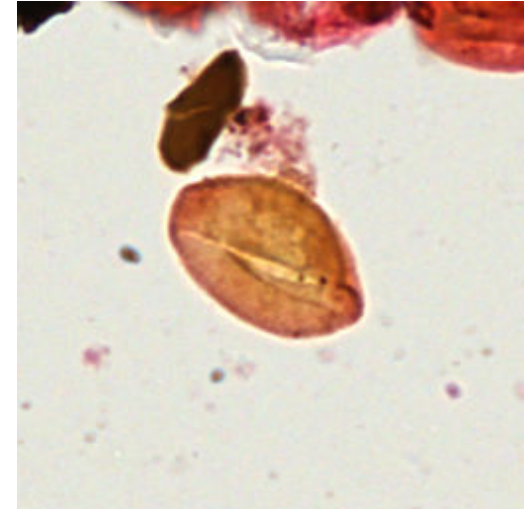
Model: Momipites sp



EXPERT LABEL

Bombacacidites sp.

Model: Rousea sp.



EXPERT LABEL

Monocolpites sp.

Model: Arecipites sp.

Selected validation mistakes—not proposed changes to the expert labels.

Help us prioritize the next improvements

- 01 Which category confusions matter most scientifically?
- 02 Which categories should we improve first?
- 03 Can we review a small set of confusing specimens together?
- 04 When is more than one focal view essential?

Next: expert error review → targeted improvements → final evaluation